

My Security Incident Report

The protocol associated with this incident is the Hypertext Transfer Protocol (HTTP). Because the problem involved accessing the web server of yummyrecipesforme.com, it is clear that communication with the site relied on HTTP traffic. When tcpdump was executed while visiting the website, the captured logs confirmed that HTTP was being used to communicate with the server. The malicious file was delivered to users through HTTP at the application layer.

Several customers reported to the website's helpdesk that upon visiting the site, they were prompted to download and execute a file that supposedly provided access to new recipes. After running the file, their computers began to operate noticeably slower. At the same time, the website owner attempted to log in to the web server but discovered that access to the administrator account had been revoked.

To investigate safely, the cybersecurity analyst accessed the website within a sandbox environment to avoid affecting the organization's internal network. The analyst then used tcpdump to capture network traffic generated during interaction with the site. While browsing, the analyst was prompted to download a file claiming to offer free recipes. After downloading and executing the file, the browser redirected to a fraudulent website, greatrecipesforme.com. Considering that the administrator had been locked out, the team concluded that the attacker likely carried out a brute-force attack to gain access to the admin account and change the password. Running the malicious file ultimately compromised users' systems.

Initially, obtaining the IP address from the DNS server via the DNS protocol worked correctly. The legitimate IP address for yummyrecipesforme.com was returned, and a connection attempt was made. After sending an HTTP request, the server responded appropriately, acknowledgments were exchanged, and data transmission began. However, after waiting for approximately two minutes to complete the connection, the designated port appeared to be overloaded. A subsequent DNS request returned an incorrect IP address that resolved to the malicious domain greatrecipesforme.com. When data transmission resumed, the browser initiated an unintended HTTP GET request using HTTP/1.1, which likely corresponded to the download of the malicious file.

To reduce the risk of similar incidents, organizations should implement a strong password policy requiring complex passwords that include uppercase letters, numbers, and special characters. Enabling two-factor or multi-factor authentication (2FA/MFA) for login systems is also essential. Additionally, password reuse should be prohibited.

From a physical security standpoint, offices should be equipped with surveillance cameras, controlled access points, and other protective measures to prevent unauthorized access or potential eavesdropping that could compromise the network.